Unless stated otherwise, all rings are assumed to have the normal addition and multiplication operations.

Part 1

1. [3 pts each] Provide the definition for each of the following:

a. Integral domain

b. Onto function

c. Ring isomorphism

d. Ideal

See the notes.

2. [3 pts each] Provide an example for each of the following (do not use an example from problems 1-15):

a. A ring without identity

The ring of even integers does not have a multiplicative identity since 1 is not even. Subrings of rings with identity provide examples such as $n\mathbb{Z}$ for $n > 1$.

b. A non-commutative ring

The ring of 2×2 matrices with real entries (or with integer, integer mod$n$, rational, or complex entries).

c. A field

A few of the obvious ones: $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{C}$.

d. A function which is not 1-1.

$f(x) = 1$ (or any constant function) and $f(x) = x^2$ for $f\colon \mathbb{R} \to \mathbb{R}$; or $f(a) = a$ and $f(a) = b$ for $f\colon \{a\} \to \{a, b\}$.

e. An onto function

$f(x) = x$ (the identity function) and $f(x) = x^3$ for $f\colon \mathbb{R} \to \mathbb{R}$; or $f(a) = a$ for $f\colon \{a\} \to \{a\}$.

f. Two rings which are isomorphic.

$\mathbb{R} \times \mathbb{R} \cong \mathbb{C}$, $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$, and of course any ring R is isomorphic to itself, $R \cong R$.

g. Two rings which are not isomorphic.

$\mathbb{R} \ncong \mathbb{C}$, $\mathbb{R} \ncong \mathbb{Z}$, $\mathbb{Q} \ncong \mathbb{Z}$, $\mathbb{Z}_3 \ncong \mathbb{Z}_4$, etc. It is easy to see that any two rings with different order are not isomorphic.

h. An ideal.

$2\mathbb{Z}$ or $n\mathbb{Z}$ for any $n > 1$; for and ring R, 0_R and R are ideals.

Part 2

[9 pts each] Choose the indicated number of problems from each section.

Rings: choose any three.

1. Complete the following:

a. Give an example of a subset of the integers which is not a ring (under the usual addition and multiplication).

The set of odd integers, $\mathbb{Z} - \{0\}$, $\{1, 2, 3, 4\}$, etc.

b. Give an example of a subset of the integers which is a ring.

$2\mathbb{Z}$ (the set of even integers), $n\mathbb{Z}$, $\{0\}$, $\mathbb{Z}$ (I didn't specify that it had to be a proper subset), etc.

2. Prove that $\mathbb{Q}$ is subring of $\mathbb{R}$.

Proof. $\mathbb{Q} \subseteq \mathbb{R}$ and $\mathbb{Q} \neq \emptyset$ since $0 \in \mathbb{Q}$ (and many others). Let $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ (so $a, b, c, d \in \mathbb{Z}$). $\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} \in \mathbb{Q}$ and $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}$. So $\mathbb{Q}$ is a subring by the subring test. $\square$

3. Complete the following:

a. Prove that the set of odd integers is not a subring of $\mathbb{Z}$.

Proof. 3 and 5 are odd integers, but $3 + 5 = 8 = 2 \cdot 4$ is even. So the set of odd integers is not a subring of $\mathbb{Z}$ since it is not closed under addition. $\square$

b. Prove that in general the union of any two rings is not a ring.

Proof. This is equivalent to showing that there are two rings such that their union is not a ring. $\{0, 1\}$ and $3\mathbb{Z}$ are subrings of $\mathbb{Z}$, but $1 + 3 = 4 \notin \{0, 1\} \cup 3\mathbb{Z}$. $M_{2\times2}(\mathbb{R})$ and $M_{3\times3}(\mathbb{R})$ are both rings, but their union certainly is not -addition an multiplication between their elements isn't even defined. $\square$

4. Prove that the intersection of any two subrings of a ring R is also a subring of R.

Proof. Let A and B be rings. Since $0_R \in A \cap B$; $A \cap B \neq \emptyset$ and we can choose $x, y \in A \cap B$. Since $x, y \in A$ and A is a ring, we have that $x - y \in A$ and $xy \in A$. Similarly $x - y \in B$ and $xy \in B$. Thus $x - y, xy \in A \cap B$, and $A \cap B$ is a subring by the subring test. $\square$

5. Let R be a ring and $a, b \in R$. Prove that $(-a)(-b) = ab$. Hint: $(a + (-a))(-b) = 0$.

Proof. Let $a, b \in R$. Using the hint, $(a + (-a))(-b) = 0 \Rightarrow a(-b) + (-a)(-b) = 0$. So $(-a)(-b)$ is the unique additive inverse of $a(-b)$ which is ab. (Since $a(-b) + ab = a(-b + b) = 0$). $\square$

Proof. $(-a)(-b) = (-a)(-b) + (-a)b + (ab) = (-a)(-b + b) + (ab) = ab$. $\square$

Integral Domains and Fields: choose any two.

6. Prove that $\mathbb{Z}_n$ is <u>not</u> an integral domain if n is not prime.

Proof. Since n is not prime, $n = mq$ for two non-zero integers such that $1 < m, q < n$. $m \bmod n$ and $q \bmod n$ are not zero, but $mq \bmod n = n \bmod n = 0 \bmod n$. $\square$

7. Prove that $\mathbb{Z}_3$ is a field.

Proof. 3 is prime so then $\mathbb{Z}_3$ is a field. $\square$

Proof. We know that is $\mathbb{Z}_3$ is a ring. $1 \cdot 1 = 1$ and $2 \cdot 2 = 1 \bmod 3$. So every element has a multiplicative inverse and $\mathbb{Z}_3$ must be a ring. $\square$

8. Let R be an integral domain and suppose that $ax - bx = 0$. Prove that if $a \neq b$ then $x = 0$.

Proof. If $ax - bx = 0$ then $(a - b)x = 0$. Since $a \neq b$, $a - b \neq 0$. So $x = 0$ since R is an integral domain. $\square$

9. Let R be an integral domain and let $a, b, c \in R$ where $c \neq 0$. If $ac = bc$ then $a = b$.

Proof. $ac = bc \Rightarrow ac - bc = 0 \Rightarrow (a - b)c = 0$. Because R is an integral domain and $c \neq 0$, $a - b$ must be zero. This implies that $a - b$. $\square$

10. Let R be a field. Prove that R is an integral domain.

Proof. Suppose that $ab = 0$ and $a \neq 0$ for some $a, b \in R$. Since R is a field, there exists an a^{-1} such that $a^{-1}ab = a^{-1}0 \Rightarrow b = 0$. So then R is an integral domain. $\square$

Homomorphisms, isomorphisms, and ideals: choose any three.

11. Prove that $f\colon \mathbb{R} \to \mathbb{R}$ where $f(x) = e^x$ is one-to-one but is not onto.

Proof. (1-1) Let $a, b \in \mathbb{R}$, $e^a = e^b \Rightarrow \ln(e^a) = \ln(e^b) \Rightarrow a = b$.
(not onto) $-1 \in \mathbb{R}$ but there is no real number such that $e^x = -1$. $\square$

12. Let $f\colon R \to S$ and $g\colon S \to T$ for rings R, S, and T. Prove that $f \circ g\colon R \to T$ is a homomorphism.

Proof. $f(g(a + b)) = f(g(a) + g(b)) = f(g(a)) + f(g(b))$, and $f(g(ab)) = f(g(a)g(b)) = f(g(a))f(g(b))$ since g and f are each homomorphisms. $\square$

13. Let $f\colon \mathbb{Z}_4 \to \mathbb{Z}_4$.

- **a.** If $f(x) = 2x$ prove that f is not a homomorphism.
- **b.** If $f(x) = 3x$ prove that f is a homomorphism.

14. Prove that $f\colon \sqrt{2}\mathbb{Z} \to \sqrt{2}\mathbb{Z}$ where $f(a + b\sqrt{2}) = a - b\sqrt{2}$ is an isomorphism ($\sqrt{2}\mathbb{Z} = \{a + b\sqrt{2}\colon a, b \in \mathbb{Z}\}$).

15. Show that any ring isomorphism $f\colon \mathbb{Z} \to \mathbb{Z}$ must be the identity map.

16. Let A and B be ideals of R. Prove that $A \cap B$ is an ideal of R.

17. Prove that $\mathbb{R} \times \mathbb{R} = \{(a, b)\colon a, b \in \mathbb{R}\}$ is isomorphic to $\mathbb{C} = \{a + b\sqrt{-1}\colon a, b \in \mathbb{R}\}$ if multiplication in $\mathbb{R} \times \mathbb{R}$ is defined as $(a, b)(c, d) = (ac - bd, ad + bc)$ and addition is defined as $(a, b) + (c, d) = (a + c, b + d)$.